

# ActInf Textbook Group ~ Cohort 2 ~ Meeting 18 (Chapter 8, part 1)

*TextbookGroup/ParrPezzuloFriston2022/Cohort\_2 | Cohort 2 ~ Meeting 18 (Chapter 8, part 1) | 53:31*

Hello, it's March 1st, 2023. We're in Cohort 2 of the Paradol textbook group. We're in Meeting 18, our first discussion on Chapter 8. First, Michael, if you wanted to just raise the context or question that you wanted to begin with, and we'll just pick up there.

You're on mute. I'm not sure if you're asking.

Sorry, my microphone was off. Yes. Yeah. So the map territory fallacy, I was wondering what this is.

And whether you could perhaps elaborate a little bit on that fallacy.

All right. I'm moving it to Chapter 9, even though we're in Chapter 8, because we're going to bring in a figure from Chapter 9. Oli, you want to give a first thought on this?

Yeah, sure. Actually, map territory fallacy is one of the fallacies that is widely used to somehow criticize a free energy principle, because as the name of the concept may suggest, some believe that it's a kind of fallacy to somehow conflate the map with the territory. So by this, I mean that free energy principle claims that, well, we have, so very informally, we have a map of the principle or the theory or theory or whatever, and the territory that it applies to. So the system or the agent that we're trying to model. But on the other hand, by claiming that the system or the agent actually does the kind of inference that recovers the map itself, some people believe that this kind of conflation that map converts into territory or the vice versa is a fallacious claim. So we need to separate map from the territory. But in this recent paper on the map territory fallacy fallacy, Ramsted et al argued that actually, this is not a fallacy at all. It is the structure of these of the the FEP theory. So in other words, we're well aware that there is a kind of inference as if epistemology happening here. So yeah, but it's a bit more extensive. They have lined up some number of arguments to defeats this map territory fallacy fallacy fallacy, and why it's not fallacious at all. So yeah, that's basically the gist of it, in my opinion.

Does it have to do with a distinction between model and process? And the model is the, maybe the model is the map and the, no, the other way, the process is the map, and the model in the external world, I understand.

No, the external world is the, is the, what is the process is the external world and the model is the, I mean, the generative model of the agent itself. Yeah, it definitely relates to that. But on the other hand, it also relates to this concept of self-evidencing, which is a way to describe the mechanism by which the model in the environment, or let's say the agent in the environment, models itself according to the generative model. So it's a kind of, as the name suggests, a self-referential way of modeling the system and the environment.

Yes, it's definitely a roller coaster, and not a resolved one. But still, we can look to some of these key recent works. So, the map territory fallacy would be just like it's been since the classical Borges writing. This realization, like, the map is not the territory, kind of related, like all models are useful, or some models are useful, but none are true. Like, correlation is not causation. This sort of broad

space of aphorisms that's basically like, instrumentalism isn't realism, as we might say it.

So that's the map territory fallacy. We'd be making the fallacy that the map is the territory. I made a linear regression between cancer and heart disease, ergo they are actually causal that way. And when we're using active inference framework, one might be slightly more tempted than the linear regression to say, well, because I've generated a model that has a structure that seems to articulate certain natural things, this is like an explanation. It might even have predictive power. And so that might lead somebody to make this kind of a fallacy that the map is the territory. Then Ramsted et al have kind of agreed and amplified, which is saying, actually, when you use the map territory fallacy to criticize the kind of cognitive modeling in the FEP and Bayesian mechanics, you're acting fallaciously. And there's several angles on that. Unpack more on the paper. One angle is you might be chopping off more than you want to. Like, you may realize it or not, your critique may extend to simply all quantitative modeling. And many people want to like, excise FEP from the space of scientific modeling in good graces. But then they don't want to, for example, dismantle the architecture of using statistics.

So that's one angle, which is like, if you actually seek out the root of your descent with this kind of cognitive modeling, you're going to be cutting off a lot of what is just understood to be statistical modeling. So it's like, you've gone so far into denying the map territory relationship that you've actually biopsied more than just the FEP.

So that's one angle. And then they go further, which is that several features of the FEP make it the ideal model of generic systems in statistical physics, which is a bit of an abstract statement. So we can bring it back to the textbook with figure 9.1, which just as Michael brought up, this came up in the Par recent book stream with Par, where we asked what was happening in this figure and had some discussion on it. And here's the experimenter in the outer shell, making a cognitive model that cognitive model of, let's just say the rat and the teammates, has an aboutness that reflects the experimental setting and context.

So it's making the experimenter, human in this outer shell, is making a cognitive model, which is a map. So all the caveats of linear regressions apply. Map's not the territory. Don't want to fall into the map territory fallacy. However, it's also the case that that cognitive model, that map, is map-like in its relationality to the experimental context. So there's this view from the outside, black box, cognitive model being proposed as a hypothesis, discrete time, chapter 7, continuous time, chapter 8, whatever it may be. But also the aboutness of that map is about this generative process, which is the generative process handing stimuli to the generative model of the agent. And so that duality, view from the inside and view from the outside, makes the FEP uniquely the ideal model of generic systems.

So they're saying, yeah, for sure, don't do the naive fallacy, but don't engage in the fallacy fallacy.

And then one final piece is, if anyone has a dissent with this, let's just see their alternative.

Let's not simply tolerate a standalone criticism, which is actually like a space for improvement.

But is anyone seriously proposing an alternative to this structure?

And the answer is not really. Ali?

I think I just to add slightly more detail to what you just explained in terms of the paper itself.

Well, they actually lay out two lines of arguments against map territory fallacy.

One is, as I mentioned, the misconception that the process and the model or the map and the territory are conflated,

which they say they're not actually conflated, but the territory behaves as if it were a map.

So it's a kind of nested representation.

And the other line of argument is, as Daniel mentioned, arises from a kind of misunderstanding of the kind of modeling used in statistical physics,

and in particular, the principle of maximum entropy to model the environment.

So the way that, I mean, in FEP, the way that agent models or self-evidences the environment is through CMAP or a constrained maximum entropy principle.

So, yeah, I just wanted to add those two bits of detail there.

Nice.

Michael?

So isn't this that a map and a territory implies that there is an isometry between the map and the territory, but in free energy, in certain and several points, they claim these things are independent.

So the model, the process and the model is actually independent through these Markov blankets and so on.

And we cannot actually know what happens outside the world, which is quite different from that idea of map and territory.

Well, great, great questions.

I think there's two threads there that are very tightly intertwined, but they're really important to consider separately.

So first, you said, is there an isomorphism or an isometry between the map and the territory?

Isomorphism in the strict sense, meaning identical structure, is not required.

The generative process, even just talking about the map, the generative process could be a continuous variable,

and the generative model could be like discrete or a binary categorical variable.

So isometry is not even required between the internal states and the external states within the map.

And then it's an even broader space where you don't even know if isometry even applies to, for example, the structure of the real territory.

So for sure, isometry or isomorphism is not required.

Some kind of adaptive resemblance will come to be seen.

That's like the good regulator theorem from cybernetics.

Any regulatory adaptive system is as if that epistemology that Ollie mentioned,

it's as if it has an effective model of its niche.

Again, that doesn't need to be isomorphic.

But evolution or far from equilibrium thermodynamics does make it the case that you're going to get swept off the table if you can't adapt.

And so over generations and through development, it'll come to be the case that there's functional relationships and a semiotic process,

but it doesn't have to be as stringent as an isomorphism.

And the second aspect was the statistical independence that is the technical definition of a Markov blanket.

So Markov blanket, it's the nodes upon which two other sets of nodes in a Bayesian graph are conditionally independent.

So the fact that we're invoking the Bayes graph right off the bat, we know that we're talking about the map.

So territories don't have Markov blankets.

Maps do.

So already we're within the map discussion space.

And what the Markov blanket does is it precludes direct consequences, direct causal impact from internal and external states.

So no telepathy, no telekinesis.

It's just a model constraint.

However, still through the blanket, there can be the propagation of causal influences on a Bayes graph.

For example, a light could turn on in the generative process that could shine photons onto the sense state, and then that could update the beliefs of the internal states.

So implications can still propagate on a Bayes graph through the conditional independence of the Markov blanket,

just like person A could tell person B, and then person B could tell person C.

And that doesn't mean that person A told C.

Yes, Michael, go for it.

So you're talking about the free energy principle.

But does this apply also to the notion of maps and territory?

As for instance, we, I mean, Google Maps, for instance, they should have a more strict notion of isometry, I guess.

It's just a question of what you were talking about, about the free energy principle or the notion of maps and territory in general.

So to what extent does this metaphor, you think that this metaphor really applies to arbitrary depths?

If I make my question clear.

I'm just, for those who are watching the video, I just have to like, my screen is like frozen.

So I'm just needing to kind of reset the screen share.

But I mean, arguably, Google Maps, it depends by what you mean by an isometry.

That's a two-dimensional GPS landscape.

So some kind of distorted representation, like making it flat, is done within the bounds of what's useful.

I'm not sure what else needs to be claimed for the purposes of Google Maps.

Okay, and so you're saying the same also applies to these Markov blankets and the free energy principle.

What do you think, Ali?

Well, again, we go back again and again to Peirce's taxonomy of symbols.

Actually, maps, so either, I don't know, Google Maps or circuit diagrams or these kinds of maps can act as a kind of icon for the object instead of the index.

So, sorry, I thought I got disconnected.

Icon instead of index.

So, an icon is a kind of sign that stands for the object through the relationship.

Okay?

So, this kind of relation between the icon and the object is a kind of structure-preserving mapping, especially in the case of geographical maps or any other cartographical different mappings.

But, in my opinion, when we talk about maps and territories in relation with physical systems or biological systems or whatever,

we use this term in a slightly more broader and a much more looser term.

And it doesn't require to be an exactly structure-preserving mapping.

So, it's more like a mathematical mapping between a mathematical relation between a sets of object or sets of states and the other sets of states,

namely generative processes and the generative model.

So, that's it.

I mean, there isn't any need to impose any additional requirement for the exact structure of those mappings.

It's a general mathematical mapping, per se.

Awesome.

Yeah.

One other piece, like, from the empirical practice side that I think informs our philosophy of science is, very rarely is the empirical researcher simply interested in producing one final model.

Modeling is almost always an iterative process that involves the maintenance and diversification and selection from portfolios of models.

So, the findings of a paper might be like,

this phase space that we explored of models, these structural families,

suggest that the two factors are better than the one factor,

or that short-term memory does interact with valence in this way.

So, right off the bat, empirically, we're talking about portfolios of models.

Hence, there's obviously a multiple-to-one mapping of maps to system of interest.

Ergo, it's a bit limited to try to over-interpret any map

and even get into this discussion of whether it's actually the territory.

It's not.

That's why we're doing it.

That's statistics.

That's modeling.

We have a whole host of models on our computer.

None of them are the territory.

But it's so tantalizing because, first off, unlike the linear regression, we're talking about the action perception loop, which brings this sort of tangibility within our grasp.

And also, there's this question of language with the instrumentalism and the realism being mixed together conceptually and linguistically.

Well, the agent is minimizing free energy.

Is it, though?

Or is it the modeler's generative model of the actual embodied organism?

So, somebody who knows that may be able to speak or interpret flexibly and understand that when people are talking about an organism minimizing surprise or the imperative for an organism to minimize surprise, be able to, on one hand, lean into the realism, and, on the other hand, lean into the instrumentalism.

But neither of those, I guess, is like a standalone interpretation or straightforward implication from any set of formalisms.

Let's go to chapter 8 and let's see how it plays out with continuous time.

All right.

As it's our first discussion in 8, let's scan through and then we'll come to the questions next week.

Okay.

So, everything flows, nothing stands still.

One of the cherished expressions of process philosophy and dynamicism.

Chapter 8 complements 7 by continuing the discussion of generative models.

Chapter 7 was about discrete time.

Chapter 8 is on continuous state-space models and it'll be brought up in multiple ways that continuous time models, continuous perceptual models, and continuous action models are often well-suited for proximate generative models.

Not using the term formally, but just like dealing most proximally with incoming sensory data and on the action side,

dealing most proximally with motor actuation.

Turns out that those

settings

often are

continuous in nature.

Like the amount of pressure

on a proprio receptor

or the amount of volume

by an auditory sensor

on the sensory sides

and then on the action side,

like the angle of an elbow

or the amount of force being exerted by a muscle,

those

perceptual and motor

settings

are really amenable to continuous modeling,

whereas

discrete time models

are often amenable to categorical,

symbolic,

and discrete

settings.

Section 8.2

and just raise your hand if you have like a

want to add something or a question.

8.2

From chapter 4,

the generative model

in continuous time

can be written as a pair

of stochastic equations

determining how states,

hidden states,

$x$ ,

generate data,

observables,

$y$ .

So that's the top equation,  
is  
the data are being generated  
by  
a generator function  
on  $x$ ,  
hidden states,  
plus a noise term  
on  $y$ ,  
and  
change in  $x$ ,  
 $\dot{x}$   
derivative on  $x$   
is a function of  
the current state of  $x$   
and  $v$ ,  
which are  
slowly varying  
causes  
or contexts,  
plus  
a noise term  
on  $x$ .

This  
is a  
classical  
formalization,  
and in the context  
of SPM  
and brain imaging,  
this is like  
 $y$  is the fMRI  
or the EEG  
measurements.

$x$   
are the  
unobserved  
neural states,



neural activity dynamics,  
and there are some  
sensor  
noise.  
And then  
the dynamics  
of neural activity,  
which again are unobserved,  
inferred from data,  
are a function of  
current state of those  
neural activity patterns,  
and slowly varying  
causes,  
which might be like  
an attentional factor  
or might be an  
experimental factor.

And more broadly,  
just this partition  
into like  
signal and noise  
comes up again  
and again.

Ali?

I just  
wanted to point  
out the  
second equation  
is also called  
Edo's  
stochastic  
differential equation.

So  
if someone  
come across  
that  
term,

that's exactly  
this  
second equation  
of  
8.1.

Awesome.

These  
stochastic  
calculi  
help us  
do  
integrals  
and derivatives  
on  
stochastic  
processes.

Action  
is absent  
from equation  
8.1.

This is because,  
as outlined  
in the recipe,  
action is part  
of the generative  
process,  
not the generative  
model.

So  
this is  
potentially like  
saying  
the consequences  
of action  
are part  
of how  
things are  
actually

generated,  
which is to  
say the  
generative  
process.  
Even  
though  
action  
selection  
is a  
function  
of a  
generative  
model,  
the  
consequences  
of action  
in terms  
of modifying  
how the  
world  
changes  
are  
occurring  
in the  
generative  
process.  
If we  
write down  
the dynamics  
of the  
real world  
generative  
process,  
but of course  
this is like  
the dynamics  
of the

map  
of the  
real world,  
we would  
have to  
include  
action.  
So we  
have a  
very  
similar  
equation,  
but  $U$   
is now  
replaced  
for  $V$   
and there's  
bold  $G$   
and  $F$ .  
this is a  
general  
formalization  
with this  
sort of  
observable  
layer  
caused by  
hidden  
states  
and the  
change in  
the hidden  
states  
being related  
to their  
current  
state  
and

causal  
factors.

So  
they're  
going to  
specify  
in  
equation  
8.3  
a  
simple  
version  
that's  
compatible  
with the  
above  
and  
noiseless.

So it's  
a point  
attractor.

When  $X$   
is less  
than  $V$ ,  
the expected  
rate of  $X$   
is positive  
and vice  
versa.

So this  
is a  
set point  
attractor.

They're  
going to  
define  
a  
generative

process  
where  
action  
will  
change  
to  
fulfill  
the  
predictions  
of  
8.3.  
So  
this is  
an  
active  
movement  
towards  
a  
point  
attractor.  
So  
that  
point  
attractor  
might  
reflect  
a  
preferred  
target  
for  
the  
eyes  
or  
a  
preferred  
location  
in  
space

for  
a  
limb  
and  
this  
is  
bringing  
us  
from  
this  
totally  
general  
setting  
into  
an  
active  
attainment  
or an  
active  
agency  
exertion  
that's  
compatible  
with those  
formalizations.

they're  
going to  
take it  
one  
level  
more  
complex.

A more  
sophisticated  
model recognizes  
that forces  
generated by  
muscles

change  
velocity  
not the  
position.  
So you  
don't just  
pick up  
and move  
your limb  
you're  
actually  
modifying  
a derivative  
of position  
which is  
velocity  
and  
acceleration.  
and so  
here  
we start  
to see  
a little  
bit  
of  
what  
could  
be  
extended  
into  
higher  
and  
higher  
derivatives  
with the  
generalized  
coordinates  
of



motion.

But now

$f$

of  $x$

and  $v$

which is

to say

the

function

that is

a function

of

the current

state

of the

hidden

state

and

the

slowly

causal

varying

factors

are

going to

be

explicitly

written

as

this.

$x_1$

is the

position

and  $x_2$

is the

velocity.

Now

this

generative  
model  
is  
describing  
Hooke's  
law.

So this  
is a  
spring  
movement  
with  
a  
conservative  
spring.

We can  
write down  
a generative  
model that  
predicts the  
dynamics that  
wouldn't fold  
if there  
was a  
spring  
drawing a  
limb to  
a  
desired  
location.

So  
that's  
like  
one  
level  
more  
advanced  
than  
the

point  
attractor  
which  
just  
linearly  
picks up  
the arm  
and moves  
it to  
the  
point.

Now  
we have  
like  
this  
Hooke's  
law  
spring  
force  
drawing  
and  
oscillating  
and  
then we  
can imagine  
you can  
keep on  
going.

You can  
have a  
damping  
spring  
and so  
on.

8.3  
Continuous  
time  
formalizations

are well  
suited to  
characterization  
of movements  
and  
really  
just  
nonlinear  
dynamical  
systems  
wherein  
discretization  
of time  
and space  
is efficient.

Hashtag  
livestream

52  
discretization  
algorithms  
and optimization.

Figure

8.1

we're  
returning  
to the  
spinal  
reflex  
case  
from  
chapter

5

descending  
predictions  
getting  
mashed  
up  
with

incoming  
proprioceptive  
data  
resulting  
in an  
error  
residual  
that is  
modifying  
the  
trajectory  
of  
the  
actions  
in the  
face  
space  
shown  
here.  
X  
is the  
position  
of the  
limb  
and  
V  
is where  
it's  
being  
drawn  
to.  
X  
is  
current  
position  
V  
is the  
target

location.

8.1

Precision

Attention

and

Sensory

Attenuation

We could

talk more

about it

but this

related to

a discussion

that we

had in

office hours

last week

about

the

sensory

attenuation.

Here

are

some of

our

discussions.

We

talked

about

why

it's

relevant

to

model

sensory

attenuation

which is

like the

reduction  
of  
precision  
on  
proprioception  
in order  
to  
facilitate  
action  
selection  
and  
that's  
also  
kind of  
described  
here.  
High  
precision  
sensory  
input  
you just  
get  
shut  
down  
no  
movement  
is  
executed  
it's  
catatonic.  
they're  
going to  
move into  
two  
classical  
dynamical  
systems  
one

with  
predator  
prey like  
dynamics  
lock of  
volterra  
and the  
second  
with  
chaotic  
dynamics  
lock of  
volterra  
predator  
prey model  
also called  
a  
winnerless  
competition  
equation  
8.6  
gives  
the lock  
of  
volterra  
dynamics  
and here  
on the  
top  
is  
the  
winnerless  
competition  
amongst  
plant  
herbivore  
and  
carnivore



also  
one  
could  
imagine  
this  
is  
like  
activity  
patterns  
in  
the  
fMRI  
or  
EEG  
with  
different  
brain  
regions  
and  
that's  
exactly  
how  
it's  
been  
used  
and  
here  
is  
the  
phase  
space  
this  
is  
like  
three  
lines  
plotted  
on

time  
axis  
but  
we  
can  
also  
think  
about  
that  
as  
a  
trajectory  
in  
a  
three  
dimensional  
space  
so  
some  
sort  
of  
cube  
and  
there's  
just  
one  
point  
ABC  
or  
PHC  
that  
are  
moving  
in  
this  
three  
dimensional  
space

on  
some  
trajectory  
and  
then  
here  
if  
you  
look  
at  
one  
side  
or  
the  
other  
of  
a  
cube  
you  
can  
look  
at  
how  
there's  
something  
like  
a  
slowly  
drifting  
limit  
cycle  
when  
you  
project  
down  
to  
a  
given

axis  
now  
they're  
going  
to  
move  
that  
log  
of  
Volterra  
winnerless  
competition  
predator  
prey  
dynamic  
into  
sequential  
action  
here's  
how  
sequential  
action  
has  
been  
used  
in  
eye  
blink  
conditioning  
experiments  
and  
in  
writing  
generation  
of  
sequential  
information  
patterns

and also  
in the  
PAR  
bookstream  
2.1  
there was  
a great  
discussion  
by Thomas  
so  
he said  
early  
act  
inf  
it was  
continuous  
it was  
generalized  
filtering  
and  
then  
there  
was  
the  
engagement  
with  
action  
people  
started  
thinking  
about  
how  
to  
get  
sequential  
dynamics  
emergence  
of

sequences  
from  
continuous  
dynamics  
was a  
key  
step  
towards  
purposeful  
planning  
that's  
figure  
8.3  
then  
it's  
like  
okay  
that's  
all  
good  
for  
making  
this  
kind  
of  
semi  
regularized  
scribble  
but  
what  
if  
you  
wanted  
actual  
policy  
selection  
on  
which

word  
to  
write  
that's  
going  
to  
be  
really  
hard  
to  
bake  
into  
a  
lock  
of  
ulterra  
model  
so  
things  
moved  
into  
the  
explicit  
discrete  
state  
space  
model  
with  
the  
POMDP  
formalization  
and that  
led to  
planning  
as  
inference  
and  
all

these  
other  
features  
and  
now  
we're  
moving  
back  
into  
these  
hybrid  
and  
hierarchical  
models  
where  
continuous  
state  
spaces  
are  
reintroduced  
as  
a  
low  
level  
of  
hierarchical  
modeling

## 8.2

learning  
continuous  
models  
continuous  
state  
spaces  
are a  
lot  
like  
discrete



state  
spaces  
in  
discrete  
state  
spaces  
you're  
basically  
taking  
a  
sum  
across  
a  
finite  
number  
of  
entities  
in  
continuous  
state  
spaces  
you're  
taking  
an  
integral  
from  
here  
to  
there  
and  
there  
are  
different  
approximations  
that  
facilitate  
those  
integrals

the  
PMDP  
formalization  
of  
chapter  
7  
has  
largely  
superseded  
the  
use  
of  
generalized  
Locke  
Volterra  
systems  
in  
Act  
Inf  
but  
it's  
still  
extremely  
didactic  
and  
pedagogical

And  
used  
in  
the  
lower  
levels  
of  
hierarchical  
models  
now  
they're

going  
to  
move  
from  
Locke  
Volterra  
to  
the  
Lorenz  
!  
Attractor  
system  
which  
has  
some  
parameters  
that  
are  
reflective  
of  
weather  
systems  
That's  
been  
studied  
in  
the  
context  
of  
birdsong  
and  
they're  
going  
to  
use  
the  
Lorenz  
attractor

in  
the  
birdsong  
case  
to  
talk  
about  
generalized  
synchrony  
So  
generalized  
synchrony  
doesn't  
mean  
lockstep  
synchronization  
it means  
that there  
are  
entities  
whose  
dynamics  
are  
on  
a  
manifold  
in  
other  
words  
their  
dynamics  
are  
not  
untethered  
from  
each  
other  
there's

mutual  
information  
such  
that  
they're  
co  
ambulating  
on  
a  
lower  
dimensional  
manifold  
So  
the  
birdsong  
example  
has  
been  
in  
multiple  
papers  
It's  
a  
chaotic  
attractor  
that's  
underlying  
the  
emission  
of  
birdsong  
by the  
two  
birds  
Live  
stream  
32  
is

on  
coupled  
chaotic  
systems  
So  
that's  
where  
the  
math  
are  
unpacked  
the  
most  
but  
here  
we  
can  
just  
see  
on  
the  
right  
side  
of  
figure  
8.5  
that  
after  
learning  
there's  
a  
tightening  
of the  
first  
and the  
second  
bird's  
expectations

towards  
a  
manifold  
It's  
called  
singing  
from  
the  
same  
hymn  
sheet  
Pristen's  
quite fond  
of that  
phrasing  
and  
each  
bird  
contributes  
sections  
of the  
song  
and  
there's  
some  
interesting  
connections  
back  
to  
sensory  
attenuation  
again  
Section  
8.5  
goes  
into  
hybrid  
discrete

and  
continuous  
models  
which  
we've  
basically  
been  
talking  
about  
multiple  
times  
so  
figure  
8.6  
first  
off  
note  
that  
the  
G  
should  
be  
connected  
to  
the  
pi  
then  
note  
the  
fractal  
rake  
structure  
we  
have  
the  
rake  
here  
figure



4.3  
on the  
bottom  
continuous  
time  
 $V$   
 $X$   
 $X$   
prime  
double  
prime  
and so  
on  
but  
these  
continuous  
time  
models  
lower  
half  
of  
figure  
4.3  
are  
embedded  
within  
a  
discrete  
time  
higher  
level  
A  
B  
D  
S  
T  
minus  
1

T  
T  
plus  
1  
so  
we  
see  
both  
parts  
of  
figure  
4.3  
discrete  
time  
on top  
D  
B  
A  
past  
present  
and future  
and now  
we're seeing  
it  
as  
a top  
level  
nesting  
continuous  
time  
lower  
level  
model  
and  
here  
X  
derivative  
of X

second  
derivative  
of  $X$   
key  
difference  
between  
discrete  
time  
and  
continuous  
time  
models  
Michael  
yeah  
I  
have a  
question  
what  
happens  
if  
the  
embedded  
time  
frames  
are not  
fitting  
into the  
upper  
frame  
so if  
for instance  
here  
the lower  
parts  
overlap  
with the  
states of  
the upper

part  
so you have  
a time  
frame  
and maybe  
and  
chopping up  
of time  
into  
segments  
but  
the  
segments  
overlap  
is  
there  
a  
possibility  
to  
model  
such  
kinds  
of  
things  
yeah  
few  
thoughts  
good  
question  
the  
first  
is that  
the  
continuous  
time  
model  
doesn't  
make

explicit  
predictions  
about  
the  
future  
the  
continuous  
time  
model  
is a  
lot  
like  
a  
Taylor  
series  
approximation  
so  
although  
it  
embodies  
some  
patterns  
that  
extend  
beyond  
the  
present  
but  
the  
continuous  
time  
formalization  
is  
actually  
always  
real  
time  
that's

one  
answer  
a  
more  
empirical  
answer  
is  
if  
you  
have  
a  
model  
where  
there's  
like  
sort  
of  
clashing  
time  
scales  
you're  
going  
to  
end  
up  
with  
multi  
collinearity  
in  
fitting  
the  
data  
to  
empirical  
results  
so  
you'll  
get

a  
variety  
of  
model  
fitting  
aberrations  
because  
the  
variance  
is  
going  
to  
be  
partitioned  
in  
a  
sort  
of  
discordant  
way  
so  
some  
random  
seed  
might  
partition  
a lot  
of  
the  
variance  
up  
here  
and  
another  
random  
seed  
partitions  
it

down  
here  
so  
if  
there's  
discordant  
overlap  
you're  
just  
going  
to  
end  
up  
with  
a  
headache  
trying  
to  
get  
that  
model  
to  
be  
useful  
so  
no  
one's  
going  
to  
stop  
you  
but  
that  
doesn't  
mean  
that  
it's  
going



to  
be  
effective  
architecture  
for  
actual  
parameterization  
does  
does  
it  
mean  
we  
need  
to  
squeeze  
somehow  
the  
stuff  
into  
these  
boxes  
into  
these  
rectangles  
embedded  
rectangles  
there's  
no  
not  
really  
a  
way  
to  
have  
these  
overlapping  
boxes  
there

what's  
the  
case  
that  
we're  
thinking  
about  
because  
pretty  
much  
here  
is  
the  
movement  
of  
time  
at  
some  
click  
and  
then  
this  
could be  
every  
second  
is  
when  
the  
new  
limb  
location  
is  
chosen  
and  
then  
within  
each  
second

there's  
the  
continuous  
movement  
of  
the  
limb  
yeah  
so  
for  
instance  
they talk  
about  
sentence  
so  
we  
have  
words  
and  
sentences  
paragraphs  
texts  
and so  
on  
but  
now  
if  
you  
look  
at  
prosodic  
features  
they  
may  
overlap  
with  
sentence  
boundaries

so  
you  
wouldn't  
have  
a  
kind  
of  
pore  
structure  
that  
fits  
into  
the  
sentence  
structure  
there  
could  
be  
two  
different  
ways  
of  
chopping  
up  
the  
sentence  
in  
an  
acoustic  
way  
and  
well  
in  
terms  
of  
words  
and  
sentences

and so  
on  
I  
think  
it's  
a  
super  
interesting  
question  
which  
is  
like  
this  
strict  
nesting  
like  
well  
icicades  
continuously  
let's  
just  
say  
and  
they're  
recognizing  
letters  
okay  
letters  
are  
strictly  
nested  
within  
words  
words  
are  
strictly  
nested  
within

sentences  
paragraphs  
and so  
on  
and one  
might  
want  
to go  
oh  
and  
paragraphs  
are  
strictly  
nested  
within  
narratives  
it's  
like  
well  
no  
not  
necessarily  
there  
might  
be  
a  
narrative  
boundaries  
that  
are  
up  
down  
left  
right  
of  
the  
paragraphs  
so

let's  
see  
it  
start  
by  
just  
graphing  
it  
how  
you  
see  
it  
and  
then  
let's  
go  
from  
there  
but  
in  
principle  
it's  
not  
going  
to  
be  
addressable  
but  
when  
you  
actually  
write  
it  
out  
you  
may  
discover  
again

a  
portfolio  
of  
model  
structures  
which  
can  
be  
used  
empirically  
to  
differentiate  
hypotheses  
and  
their  
maps  
so  
maybe  
it  
turns  
out  
that  
a  
simpler  
model  
where  
prosody  
is  
sentence  
delimited  
is  
more  
effective  
because  
there's  
so  
fewer  
parameters



so  
the  
Bayesian  
information  
criterion  
favors  
that  
model  
but  
that  
doesn't  
mean  
that  
prosody  
is  
sentence  
delimited  
it  
means  
that  
a  
statistical  
structure  
with  
sentence  
delimited  
prosody  
was  
selected  
for  
information  
theoretic  
purposes  
map  
territory  
but  
we  
want

to  
say  
things  
about  
the  
territory  
so  
that's  
always  
the  
challenge  
we  
do  
a  
regression  
between  
ant  
foraging  
activity  
and  
the  
humidity  
and  
we  
want  
to  
say  
when  
it's  
more  
humid  
the  
ants  
forage  
more  
that  
seems  
like

the  
natural  
thing  
to  
say  
and  
it  
might  
be  
our  
motivating  
question  
but  
it's  
always  
hard  
to  
remember  
that  
it's  
like  
higher  
values  
in  
the  
humidity  
measure  
are  
correlated  
with  
higher  
foraging  
activity  
it's  
like  
just  
say  
that

foraging  
increases  
with  
more  
humidity  
fallacy  
but  
that  
is  
the  
departure  
point  
of  
the  
map  
territory  
fallacy  
so  
it's  
often  
more  
resolvable  
than  
people  
want  
to  
know  
in  
the  
specifics  
that's  
the  
deflationary  
perspective  
and  
less  
resolvable  
than

we want  
to  
know  
in  
principle  
and  
also  
one  
reason  
behind  
formulating  
sorry  
the  
continuous  
dynamics  
in  
terms  
of  
Taylor  
series  
is  
because  
it  
also  
maps  
onto  
the  
Lagrangian  
formulation  
of  
mechanics  
because  
one  
of the  
reasons  
Lagrangian  
mechanics  
is used

instead  
of the  
traditional  
Newtonian  
mechanics  
is to  
just  
get rid  
of  
time  
parameters  
so  
instead  
of  
time  
they  
use  
momentum  
and  
also  
the  
Lagrangian  
so  
we  
don't  
need  
time  
explicitly  
defined  
or  
described  
in  
order  
to  
account  
for  
the  
behavior

of  
the  
trajectories  
or  
to  
describe  
the  
behaviors  
of  
the  
trajectories  
what  
all  
we  
need  
is  
this  
series  
of  
success  
differentiations  
in the  
terms  
of  
Taylor  
series  
to be  
able  
to  
describe  
the  
whole  
behavior  
of  
the  
system  
because  
accounting

explicitly  
for  
time  
parameter  
is  
much  
difficult  
to  
model  
than  
describing  
these  
dynamical  
systems  
in terms  
of  
their  
generalized  
coordinates  
in other  
words  
their  
Lagrangian  
formulations  
and  
in the  
path  
integral  
formulation  
of  
FEP  
that's  
basically  
the  
solution  
proposed  
by  
Dalton



Friston  
and others  
to  
use  
Lagrangian  
formulation  
to  
describe  
these  
systems  
nice  
good  
point  
equation  
8.8  
is  
describing  
the  
transformation  
from  
the  
discrete  
target  
selection  
of  
the  
upper  
level  
of  
this  
model  
and  
the  
way  
that  
that  
is  
received

by  
the  
continuous  
time  
model  
so  
here  
the  
discrete  
time  
target  
selection  
is  
then  
received  
by  
the  
continuous  
time  
i  
saccade  
model  
that  
is  
an  
experimental  
setting  
that  
has  
been  
studied  
relatively  
extensively  
here  
in the  
middle  
of  
figure

8.7

we

see

the

target

selection

being

modeled

and

the

circadian

and

here's

the

calling

back

to

that

spring

like

movement

here's

like

the

acceleration

and

the

dampening

as

the

target

is

fixed

upon

so

eye

circadian

has

been  
studied  
relatively  
extensively  
in  
the  
context  
of  
information  
driven  
continuous  
modeling  
box  
8.3  
is  
a bit  
of a  
side  
note  
on  
discretization  
so  
a  
Gaussian  
mixture  
model  
or  
very  
related  
is  
a  
k  
means  
model  
where  
one  
takes  
a

cluster  
of  
data  
points  
in  
space  
and  
asks  
how  
many  
clusters  
k  
fit  
this  
scattering  
of  
points  
you  
could  
just  
have  
one  
k  
with  
just  
a  
point  
in  
the  
center  
and  
then  
there's  
some  
variance  
distribution  
around  
it

or  
you  
could  
say  
2k  
and  
then  
you  
can  
look  
at  
the  
error  
profiles  
to  
determine  
adequate  
number  
of  
clusters  
somewhere  
between  
one  
cluster  
and  
one  
cluster  
per  
data  
point  
there's  
going  
to be  
a  
k  
means  
value  
that

is  
like  
statistically  
optimal  
8.6  
in this  
chapter  
which is  
relatively  
short  
and  
in  
some  
ways  
kind  
of  
introducing  
the  
continuous  
time  
formalism  
so  
that  
we  
can  
get  
to  
these  
important  
topics  
like  
hybrid  
models  
they  
overview  
applications  
of  
continuous

time  
GMs  
this  
is  
a  
huge  
topic  
and  
much  
has  
been  
left  
out  
see  
table  
8.1  
synthetic  
birdsong  
ocular  
motor  
delay  
conditioned  
reflex  
smooth  
pursuit  
eye  
movement  
2012  
psychosis  
illusions  
2012  
cicades  
many  
papers  
action  
observation  
attention  
hybrid



discrete  
continuous  
models  
self  
organization  
it's a  
foundation  
for  
further  
exploration  
they  
focused  
on  
movement  
prediction  
it  
simplifies  
the  
treatment  
of  
motor  
control  
we  
don't  
need  
any  
additional  
machinery  
or  
inverse  
models  
and  
this  
was  
also  
a  
topic  
picked

up  
in  
the  
skilled  
performance  
discussion  
of  
Hippolito  
et al  
talking  
about  
interactionism  
rather than  
instructionalism  
they  
talked  
they  
talked  
about  
the  
latke  
volterra  
winterless  
competition  
predator  
prey  
and the  
lorenza  
tractor  
chaotic  
system  
in all  
those  
cases  
it was  
possible  
to find  
this

concept  
of  
generalized  
synchrony  
which isn't  
lockstep  
but it's  
like a  
mutual  
information  
or  
manifold  
!  
code  
traversal  
and  
then  
they  
introduced  
the  
hybrid  
models  
which  
tend  
to have  
continuous  
lower  
level  
dynamics  
and  
discrete  
higher  
level  
dynamics  
that's  
it for  
today  
next

time  
we're  
going  
to  
come  
back  
to  
chapter  
nine  
which  
is  
going  
to  
be  
talking  
about  
model  
based  
data  
analysis  
fitting  
models  
to  
data  
and  
next  
week  
we'll  
do  
chapter  
eight  
questions  
and  
come  
through  
it  
again  
in

preview

nine

so

thank

you

fellas

see

you

next

time